Omprakash Ramalingam Rethnam

Assistant Lecturer in Construction Management · Technological University Dublin

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Academic Record

Jan 2021 to Dec 2024 – Indian Institute of Technology Bombay

- Doctor of Philosophy– Construction Technology and Management
- Final Grade: 9.18/10
- Awardee of the prestigious Prime Minister Research Fellowship (PMRF), Jan 2021 – Jan 2025, funded by the Government of India

Jul 2016 to May 2018 – Indian Institute of Technology Madras

- Master's in Technology – Construction Technology and Management
- Final Grade: 9.46/10
- Published one international journal in Life Cycle cost of 1MW solar PV power plant in BEPAM journal, an Emerald publication
- Published one conference paper on using solar energy for operating waste water treatment plants in the World Construction conference, Colombo, 2018 Srilanka and won the prestigious BEPAM Highly commended paper award

Jul 2010 to May 2014 - College of Engineering Guindy, Anna University Chennai.

- Bachelor of Engineering - Mechanical Engineering
- Final Grade: 8.59/10 – secured First Class with distinction
- Secured 'S' grade (highest grade) in Heat and Mass transfer in Semester Examination.

Jun 2009 to Mar 2010 - RSK Higher Secondary School, Kailasapuram, Trichy

- Specialization: Physics, Chemistry, Computer Science and Mathematics
- Final Mark: 97% (1162/1200), Tamil Nadu State Board–secured cent percent in Mathematics

Jun 2007 to Mar 2008 - RSK Higher Secondary School, Kailasapuram, Trichy

- Specialization: Mathematics, Science, Social Science
- Final Mark: 92% (460/500), Central Board

Academic Appointment

May 2026 – present – Technological University Dublin

Assistant Lecturer in Construction Management (Permanent Wholetime),

- Co-delivered TU822 – Building Engineering modules in building services engineering.
- Assessed the Mechanical Ventilation System Design Project for a secondary pharmaceutical manufacturing facility producing antibiotic vials for injection.
- Guided students in HVAC system design, ventilation calculations, and air distribution strategies.

Research/Teaching Experience

Jan 2025 – May 2026 – Technological University Dublin
Postdoctoral Researcher (Built Environment Research and Innovation Centre | School of Surveying and Construction Innovation)

- Working on the SEAI-funded CC-DORM project, which investigates overheating risks in dwellings under large-scale retrofit programmes and future climate change scenarios.
- Contributing to the PREDICT project within the Irish Building Stock Observatory (IBSO) team, focusing on reducing the performance gap in building energy modelling through enhanced simulation workflows and data analytics.
- Serving as Work Package Lead for policy and regulatory framework recommendations to improve building resilience and thermal comfort.

- Research areas include energy optimisation, digital construction analytics, building performance simulation, and climate-adaptive design.
- Delivered the Heating Systems and Engineering module (lectures, labs, and tutorials) for TU825 – Building Engineering (3rd year) students under the School of Mechanical Engineering.
- Delivered lectures, tutorials, and laboratory sessions on building heating systems and thermal engineering.
- Taught topics including human thermal comfort, heat transfer mechanisms, building heat load calculations, thermal capacity and intermittent heating, pump and hydronic system design, high-temperature hot water (HTHW) systems, pressurisation methods, and pipe sizing.
- Covered closed and open heating systems, centrifugal pump performance, variable-speed pump control, and the design and operation of large-scale heating networks.
- Developed teaching materials, assessments, and practical exercises related to building services engineering and HVAC system design.
- Assessed student assignments, laboratory work, and examinations.
- Supported the Geospatial Awareness module for TU835 – Planning and Environmental Management (Year 1) students in the School of Surveying and Construction Innovation.

Jan 2021 to Dec 2024 – Indian Institute of Technology Bombay

Prime Minister Research Fellow (Construction Technology and Management | Dept. of Civil Engineering)

- Conducted research in Urban and Community-scale Building Energy Modelling under the supervision of Prof. Albert Thomas, focusing on energy performance analysis, urban-scale thermal comfort, and climate-resilient design.
- Worked as an in-house Teaching Assistant, contributing to the development of teaching materials, designing examination assessments, and conducting hands-on sessions for full-semester courses that include "Construction Technology and Personnel Management" and "Construction Planning and Control."
- Delivered an expert lecture on "An Urban Modelling Perspective to Energy Efficiency and Thermal Comfort – Some Research Insights" as part of the ongoing course CVL 777 Building Sciences, Department of Civil Engineering, IIT Delhi (10 Sept 2024).
- Completed 10 hours of teaching commitment under the Prime Minister Research Fellowship (PMRF) scheme, delivering a guest lecture titled, "Introduction to Rhino and Grasshopper for Energy Modeling"– for Civil Engineering students at IIT Bombay.
- Delivered a 17-hour online crash course on Python Programming for undergraduate students pursuing a Bachelor of Computer Applications (BCA) at SONA College of Arts and Science.
- Conducted a 4-hour online session on Agent-Based Modeling for Civil Engineering students at Pillai HOC College of Engineering & Technology.
- Delivered ~100 hours (~50 hours per module) of online lectures on Python programming fundamentals and advanced for 11th-grade Chennai Higher Secondary School students (2024-PMRF teaching commitment).

Academic Awards and recognition

Feb 2026 – Seal of Excellence, European Commission – Horizon Europe
Awarded under the Marie Skłodowska-Curie Actions (MSCA Postdoctoral Fellowships 2025)
Project: "Urban Dwelling Overheating and Risk Mitigation (U-DORM)" (Proposal ID: 101277517)

Score: 94.8% (Highly ranked; above funding threshold)

Recognised as a high-quality proposal following international peer review; not funded due to budget limitations and recommended for alternative funding sources

Host Institution: Technological University Dublin

Dec 2025 – Shortlisted Nominee, TU Dublin Research & Innovation Awards 2025 (Multidisciplinary Research Team Award)
Irish Building Stock Observatory (IBSO) Team, TU Dublin
Recognised as part of the IBSO multidisciplinary team shortlisted for excellence in collaborative research and innovation (Faculty awards ceremony held 10 Dec 2025, Bolton Street Campus) https://www.tudublin.ie/research-innovation/vice-president-office-for-r-and-i/research-innovation-awards/

Jan 2025 – "Naik and Rastogi Award (Best PhD Thesis Award) for Excellence in PhD Research (2023–2025)", Indian Institute of Technology Bombay

Awarded by IIT Bombay in recognition of outstanding doctoral research performance. This institute-level honour—formerly known as the Best PhD Thesis Award—is among the most prestigious recognitions for excellence in PhD research across all disciplines.

Mar 2023: Awarded the first "R Subramanian Fellowship" award from the Glazing Society of India with Tripti Singh Rajput for the research proposal submitted- the award was given for the research proposal submitted to improve the energy performance of glazing using building energy simulation and optimization. Amount Awarded: INR 0.2 million ($ 2,403)

Dec 2020 – Selected as the awardee of the prestigious "Prime Minister Research Fellowship" for the cycle Dec 2020 – Dec 2024, funded by Government of India - the scheme seeks to attract the best talent pool of the country into research, thereby realizing the vision of development through innovation
https://dec2021.pmrf.in/

July 2018 – Highly commended paper award –7th World Construction Conference presenting "Feasibility of using solar energy to operate waste-water treatment plants" organized by Ceylon Institute of Builders, Srilanka & Department of Building Economics, University of Moratuwa, Srilanka

July 2016 – Selected for the most distinguished "Build India Scholarship" for the cycle July 2016 - May 2018, funded by Larsen & Toubro Limited for doing Masters in Construction Technology and Management at Indian Institute of Technology Madras
https://www.lntecc.com/build-india-scholarship/

Apr 2014 – First prize – Paper Presentation on "Self-Inflated Tires" - presented at Pinnacle, A national level technical symposium for Mechanical Engineers, organized by Department of Mechanical Engineering, College of Engineering Guindy.

Research Proposals (Collaborated and bagged)

Title of the Proposal: " A Data-driven Simulation-based Machine Learning Optimization SML-Opt Framework for Net Zero Energy Building Retrofits"

Lead PI(s): Dr. Albert Thomas (Indian Institute of Technology Bombay, India), Dr. Elie Azar (Carleton University, Canada)

Funding Source: DST, Government of India and IC-Impacts, Canada

Duration: 2 years (2023-2025)

Amount Awarded: INR 12 million ($ 1,44,187)

Status: Funded

My Role: Collaborated closely with Dr. Albert Thomas and Dr. Elie Azar in formulating and co-authoring a significant portion of the research proposal, contributing to the conceptual framework, literature review, and research design. Played a pivotal role in securing $1,44,187 in funding for the project through grant application support and budget development. Coordinated international research partnerships with faculty members from Canadian university, facilitating data sharing and collaborative analysis https://dst.gov.in/sites/default/files/Announcement%20of%20results%2C%20India%20Canada%20call-2022.pdf

Title of the Proposal: "Development of SIMecc-Opt framework for optimizing glazing parameters enhancing residential building thermal comfort and energy performance at minimum life cycle cost"

Award winners: Omprakash Ramalingam Rethnam and Tripti Singh Rajput

Lead PI(s): Dr. Albert Thomas (Indian Institute of Technology Bombay, India)

Funding Source: Glazing Society of India (GSI)

Duration: 2 years (2023-2025)

Amount Awarded: INR 0.2 million ($ 2,403)

Status: Funded

My Role: Won the first "R Subramanian Fellowship" award from the Glazing Society of India for recognizing the contributions so far and for researching further through the proposal on the use of energy efficient glass in the building industry. Authored the proposal along with Tripti Singh Rajput from within the SIM lab contributing to the conceptual framework literature review, and research design, grant application support and budget development https://shorturl.at/cloHK.

Submitted Research Proposals (Collaborated and Not funded)

Title of the Proposal: "Developing an open-source data-driven Multi-Agent Platform for Environment and Energy Optimization (MAPEO) toward intelligent indoor thermal comfort management"

Lead PI(s): Dr. Albert Thomas (Indian Institute of Technology Bombay, India), Dr. Ashrant Aryal (Texas A&M University, U.S.)

Funding Source: DST, Government of India

Duration: 3 years (2023-2026)

Budget proposed: INR 18 million ($ 216,175)

Status: Awaiting results

My Role: Collaborated closely with Dr. Albert Thomas and Dr. Ashrant Aryal in in formulating and co-authoring a significant portion of the research proposal to the conceptual framework, literature review, research design, grant application support and budget development.

Title of the Proposal: "Integrated framework for optimizing the net energy consumption of the building community at the least life cycle cost utilizing passive building retrofits and renewable energy shared within building community premises"

Lead PI(s): Dr. Albert Thomas (Indian Institute of Technology Bombay, India), Dr. Rakesh Kumar Mishra (Indian Institute of Technology (BHU) Varanasi

Funding Source: DST, Government of India

Duration: 2 years (2023-2025)

Budget proposed: INR 5 million ($ 60,048)

Status: Awaiting results

My Role: Collaborated closely with Dr. Albert Thomas and Dr. Rakesh Kumar Mishra in in formulating and co-authoring a significant portion of the research proposal, contributing to the conceptual framework, literature review, research design, grant application support and budget development. Coordinated inter-institute research partnerships with faculty members and research scholars from Indian Institute of Technology (BHU) Varanasi , facilitating data sharing and collaborative analysis.

Title of the Proposal: "Development of a Plastic Cash Converter Machine utilizing the power of Artificial Intelligence to create financial incentives for recycling scrap plastic": The machine will employ AI and machine learning to detect plastic types. It incentivizes users by awarding points for recycling. Optical sensors and intelligent algorithms enable efficient sorting and processing of plastic waste, revolutionizing the recycling process.

Lead PI(s): Omprakash Ramalingam Rethnam (Indian Institute of Technology Bombay, India)

Funding Source: Institute of Eminence, Indian Institute of Technology Bombay, India

Duration: 1 year (2023-2024)

Budget proposed: INR 1.1 million ($ 13,208)

Status: Not Funded

My Role: Formulating and co-authoring a significant portion of the research proposal, contributing to the conceptual framework, literature review, and research design. Coordinated with undergrad students within campus to create a conceptual model of the machine, providing the solution and value preposition, customer validation, competitive analysis, target, and revenue model, with scheduled milestones.

Title of the Proposal: "Improving the energy and thermal performance of the slum building stock using fabric solar PV integrated IOT systems".

Lead PI(s): Omprakash Ramalingam Rethnam (Indian Institute of Technology Bombay, India)

Funding Source: Tinkerers Laboratory, Indian Institute of Technology Bombay, India

Duration: 1 year (2022-2023)

Budget proposed: INR 0.31 million ($ 3,721)

Status: Not Funded

My Role: in formulating and co-authoring a significant portion of the research proposal, contributing to the conceptual framework, literature review, and research design.

Title of the Proposal: “"Development of a digital twin-based building monitoring framework to redesign the office working environment to suit COVID restrictions".

Lead PI(s): Dr. Albert Thomas (Indian Institute of Technology Bombay, India)

Funding Source: ASEAN-India Science, Technology & Innovation Cooperation, Government of India

Duration: 2 years (2022-2024)

Budget proposed: INR 1.5 million ($ 17,934)

Status: Not Funded

My Role: Collaborated closely with Dr. Albert Thomas in in formulating and co-authoring a significant portion of the research proposal, contributing to the conceptual framework, literature review, and research design. Coordinated international research partnerships with faculty members from universities from South East Asian Nations (The University of Danang - University of Science and Technology Danang, Vietnam, and Universiti Teknologi Malaysia), facilitating data sharing and collaborative analysis.

Title of the Proposal: "Decarbonizing the building sector to realize 2070 net-zero emission goals through an exploratory analysis based on development of a community-based building energy retrofit framework tool using a coupled UBEM and machine learning approach "

Lead PI(s): Dr. Albert Thomas (Indian Institute of Technology Bombay, India)

Funding Source: DST, Government of India

Duration: 3 years (2022-2025)

Budget proposed: INR 5.5 million ($ 66,085)

Status: Not Funded

My Role: Collaborated closely with Dr. Albert Thomas in in formulating and co-authoring a significant portion of the research proposal, contributing to the conceptual framework, literature review, and research design.

Journal Publications

Ramalingam Rethnam, O., Bansal, D., Thomas, A. 2026. “Satellite and Digital Image Processing (SDIP) Framework for Enhancing Community Building Energy Model (CBEM) Inputs.” Journal of Building Engineering (Elsevier)- https://doi.org/10.1016/j.jobe.2026.115830 {Q1}

Ramalingam Rethnam, O., Salunke, S., RAAJ, R., Thomas, A. 2025. “A UAV-Derived Data Processing (UDDP) framework for enhancing input reliability to create accurate community building energy models (CBEMs).” Journal of Computing in Civil Engineering (ASCE) – https://doi.org/10.1061/JCCEE5.CPENG-6440 {Q1}

Markarian, E., Qiblawi, S., Krishnan, S., Divakaran, A., Ramalingam Rethnam, O., Thomas, A., and Azar. E. . 2024. “Informing building retrofits at low computational costs: a multi-objective optimization using machine learning surrogates of building performance simulation models.” Journal of Building Performance Simulation (Taylor & Francis). https://doi.org/10.1080/19401493.2024.2384487 {Q1}

Ramalingam Rethnam, O., and Thomas, A. 2024. “A physics-informed deep learning-based Urban Building Thermal Comfort Modeling and Prediction framework for identifying thermally vulnerable building stock.” Smart and Sustainable Built Environment https://doi.org/10.1108/SASBE-02-2024-0047 {Q1}

Ramalingam Rethnam, O., and Thomas, A. 2024. “A Modeling-based Decision Support System for Enabling Mass Net-Zero Energy Retrofit of Building Communities in Developing Countries.” Journal of Architectural Engineering (ASCE) https://doi.org/10.1061/JAEIED/AEENG-1738 {Q1}

Ramalingam Rethnam, O., and A. Thomas. 2023. “A Community Building Energy Modelling – Life Cycle Cost Analysis Framework to Design and Operate Net Zero Energy Communities.” Sustainable Production and Consumption., 39 (May): 382–398. (SE: Road to Net-Zero). https://doi.org/10.1016/j.spc.2023.04.022 {Q1}

Ramalingam Rethnam, O., and A. Thomas. 2023. “Urban Building Energy Modelling-Based Framework To Analyze The Effectiveness of the Community-Wide Implementation of National Energy Conservation Codes.” Smart and Sustainable Built Environment. https://doi.org/10.1108/SASBE-09-2022-0210 {Q1}

Ramalingam Rethnam, O., Palaniappan, S., and Ashokkumar, V. 2020, "Life cycle cost analysis of 1MW power generation using roof-top solar PV panels", Built Environment Project and Asset Management, Vol. 10 No. 1, pp. 124-139. https://doi.org/10.1108/BEPAM-12-2018-0161 {Q1}

Conference Publications

Ramalingam Rethnam, O., Ahern, C., Mulville, M. . (Accepted, 2026). Climate Change–Induced Overheating Risk in Irish Dwellings: A Machine Learning–Based Assessment. 3rd Smart and Sustainable Built Environment (SASBE

2026), Cambridge, UK, 22–24 July 2026. Accepted for presentation; proceedings publication forthcoming in Springer Nature.

Ramalingam Rethnam, Mulville, M. (Accepted, 2026). A Model for Delivering Resilience and Adaptive Capacity in the Built Environment. World Sustainable Built Environment Conference (WSBE26), Melbourne, Australia, 10–12 June 2026. Accepted for presentation.

Ramalingam Rethnam, O., Ahern, C., Mulville, M. (2026). Investigating Dwelling Overheating Risks in Cooler Climates Using Ensemble-Based Machine Learning Surrogates of Parametric Dynamic Simulation Models. In: Berardi, U., António, J., Simões, N. (eds) Construction, Energy, Environment and Sustainability. ICCEES 2025. Lecture Notes in Civil Engineering, vol 744. Springer, Singapore. https://doi.org/10.1007/978-981-95-1826-5_35

Mulville, M, Harrington, Seamus., Ahern, C., and Ramalingam Rethnam, O. 2025. "Dwelling Overheating in Cool Climates: Evidence from Monitoring" CIB World Building Congress 2025, May 19th to 23rd, 2025, Purdue University, USA. https://docs.lib.purdue.edu/cib-conferences/vol1/iss1/226/

Gonuguntla, B., Rajput, T.S., Ramalingam Rethnam, O., and A. Thomas. 2024. " An efficient decision-support tool for urban building energy modeling in developing nations, leveraging satellite imagery and machine learning techniques" 2024 European Conference on Computing in Construction, July 14th to 17th, 2024, Crete, Greece. ISBN: 978-9-083451-30-5 https://ec-3.org/publications/conference/paper/?id=EC32024_244

Krishnan, S., Divakaran, A., Ramalingam Rethnam, O., Markarian, E.., Thomas, A., and Elie, A. 2024. " An efficient decision-support tool for urban building energy modeling in developing nations, leveraging satellite imagery and machine learning techniques" 2024 European Conference on Computing in Construction, July 14th to 17th, 2024, Crete, Greece.

ISBN: 978-9-083451-30-5. https://ec-3.org/publications/conference/paper/?id=EC32024_263

Bansal, D., Ramalingam Rethnam, O. and Thomas, A. 2024. " An efficient decision-support tool for urban building energy modeling in developing nations, leveraging satellite imagery and machine learning techniques" SimBuild IBPSA Conference May 21st to 23rd, 2024, Denver, Colorado. https://publications.ibpsa.org/conference/paper/?id=simbuild2024_2186

Ramalingam Rethnam, O., and Thomas, A. 2023. “A Customizable Community-Building-Energy-Modeling Decision Support System (CCBEM-DSS) for Net-Zero Planning in Developing Countries.” 2023 Winter Simulation Conference (WSC), San Antonio, TX pp. 3166-3177, doi: 10.1109/WSC60868.2023.10407443 https://ieeexplore.ieee.org/document/10407443

Ramalingam Rethnam, O., and A. Thomas. 2022. "A Community-Based Urban Building Energy Modeling Framework for Transitioning towards Zero Energy Building Communities in Developing Countries." Proc. uSIM 2022 International Building Performance Simulation Association Conference. Novemb. 25th, Scotland, United Kingdom, 1–9. https://publications.ibpsa.org/proceedings/usim/2022/papers/usim2022_p103.pdf

Ramalingam Rethnam, O., Palaniappan, S., and Pandian Ganesh Kumar. 2018, "Techno-economic feasibility of using solar energy for operating sewage treatment plants", 7th World Construction Symposium on “Built Asset Sustainability: Rethinking Design, Construction, and Operations”, Building Economics and Management Research Unit (BEMRU), pp. 527-536 (Link to read through the publication: https://ciobwcs.com/downloads/WCS2018-Proceedings.pdf

Invited Talks, Presentations & Posters

Oct 2025 – Poster Presentation, SEAI National Energy Research and Policy Conference, Ireland
Poster Title: “Climate Change-related Domestic Retrofit Overheating Risk Mitigation (CC-DORM)”
Authors: Dr Omprakash Ramalingam Rethnam; Dr Ciara Ahern (Co-PI); Dr Mark Mulville (PI)

Apr 2025 – Invited Speaker, National Construction Summit 2025 (NCS 2025), Dublin, Ireland
Talk: “Climate Change related Domestic Retrofit Overheating Risk Mitigation (CC-DORM)” (SEAI-funded project), Smart & Green Stage

Intellectual Property / Software Disclosure

CC-DORM Tool (Climate Change & Domestic Overheating Risk Mapping Tool) – Approved Software Disclosure, TU Dublin Knowledge Transfer Office, Invention ID: DIS-25-048, Technology ID T-26-005, Approved: 13 Jan 2026. Inventor contribution: 30% (with Dr. Mark Mulville and Ciara Ahern).

License(s)

July 2023: Drone Pilot License issued by the Directorate General of Civil Aviation (DGCA- Certificate No. PC072300004EV to operate small-sized rotorcraft (2 kg - 25 kg), valid for a ten-year duration

Membership in Professional Organizations

Active member of ASCE (American Society of Civil Engineers) - Member Number:000012320621

- Jointly organized a webinar titled “Precast in Mass Construction of Real Estate and Infrastructure Projects” presented by Dr. Mustafa Mashal, American Society of Civil Engineers-Indian Section Western Region (ASCE-ISWR)

Software Skills

Building Energy Modeling & Simulation:

- Grasshopper-Rhino (Honeybee, Ladybug, Dragonfly)
- IES VE (Virtual Environment)
- IES ICD (Intelligent Community Design)
- DesignBuilder
- City Energy Analyst
- EUReCA (Python Urban Building Energy Modeling tool)

Geospatial Analysis & UAV Mapping:

- Quantum GIS
- WebODM (UAV photogrammetry processing)
- Certified Drone Pilot (DGCA, India) – Licensed to operate small-sized rotorcraft (2 kg–25 kg)

Machine Learning & Deep Learning:

- Python (TensorFlow)
- Computer vision-based object detection using YOLO

Optimization & Computational Tools:

- GAMS (Optimization)
- Homer (Hybrid Renewable Energy Systems)
- NetLogo (Agent-Based Modeling)

Construction Planning & BIM Tools:

- Primavera
- Autodesk Revit
- Autodesk Navisworks
- Autodesk AutoCAD 2015

Water and Sewer Network Modelling:

- Bentley SewerGEMS
- Bentley WaterGEMS

Web Development & Programming:

- HTML
- CSS
- Python 3.x

Office Productivity:

- Microsoft Office 365

Community/Voluntary service

June 2022 to April 2024

- Website lead coordinator of CTaM department (IIT Bombay), handling the official CTaM department website using tools like GitHub, Win SCP, Atom, which has 80+ students enrolled (https://www.civil.iitb.ac.in/~ctam/).
- Strategized Plan of Action every week to update the Official department Website.
- Working in a team of 3+ coordinators, ensuring smooth operation of the updated website for entire CTaM department.

Jan 2021 to April 2022

- Organized an online free career guidance workshop called Career Reach catering mainly to the students who are in their 11th and 12th class, connecting top experts from the fields of Engineering, Medicine, Civil Services, Arts & Science, Computer Science & Information Technology to share their insights and learnings (http://surl.li/ckiwb)
- A special handy career path finder module was developed and published for school students to help students decide career opportunities that best suit their personality traits
- Published free job opportunity updates for variety of job seekers through online platform (http://surl.li/ckixa)

Professional Experience

Jun 2018 to May 2020 - Larsen and Toubro Construction, Contracts department Assistant Manager (Mechanical) (Waste-Water Business Unit).

- Responsible for complete bid preparation, including developing the Tender Information Sheet, reviewing significant contractual clauses, preparing the technical bid, electro-mechanical estimations, and compiling the risk review document up to bid submission.
- Validated offers for all process and mechanical equipment, including:
- Electro-mechanical equipment: Submersible centrifugal pumps, horizontal centrifugal/split casing pumps, aeration blowers, mechanical coarse/fine screens, sluice gates, SBR systems, chlorination systems, conventional/vortex grit mechanisms, UV systems, hydro-pneumatic surge arresting pressure vessels, centrifuges, clarifiers, thickeners, and dosing systems.
- Valves: Gate valves, butterfly valves, non-return valves, knife gate valves, ball valves.
- Pipes & Fittings: Ductile iron, cast iron, mild steel, stainless steel pipes and fittings.
- Reviewed all technical deliverables from various departments to ensure completeness and compliance prior to bid submission.
- Coordinated with international and local suppliers for quotations, performed techno-commercial comparisons for all equipment, and arrived at final estimates.
- Contributed to business development efforts by preparing technical documents and commercial justifications to educate prospective clients and facilitate new tender releases.
- Collaborated with design, operations & management, and supply chain management teams to ensure smooth project transitions by creating effective interfaces across disciplines.
- Developed and delivered technical presentations to onboard and train young engineers on the fundamentals of critical mechanical equipment.
- Showcased a special interest in renewable and sustainable construction, leading to the proposal of using solar PV panels to power sewage treatment plants — an initiative that extended to conference presentations and publications.

Jun 2014 to May 2018 - Larsen and Toubro Construction, Engineering Design and Research Centre – Senior Design Engineer- Mechanical, Utility & In-plant Piping (Waste-Water Business Unit).

- Designed sewerage networks using Bentley SewerGems for Guntur Zone II.
- Designed sewerage and raw water/clear water pumping stations for both ongoing projects and tender submissions.
- Performed pump selection based on hydraulic requirements and application specifications.
- Contributed to the design and obtained client approval for General Arrangement (GA) drawings of pumping stations, including layout and in-plant piping for the Rampur sewage treatment plant project.
- Played a key role in the Pali project (Rajasthan), contributing to the complete design process — from sewerage network design to the treated effluent pumping station during the tender stage.
- Prepared detailed Process and Instrumentation Diagrams (P&IDs) for pumping stations.
- Raised Purchase Requisitions for a wide range of equipment, including:
- Pumps: Submersible pumps, end suction pumps, horizontal split casing pumps, vertical turbine pumps.

- Screens and Gates: Mechanical and manual screens, sluice gates.
- Valves: Sluice valves, non-return valves.
- Hoists and Cranes: Manually and electrically operated.
- Piping and Fittings: Pipes and fittings for both pumping stations and sewage treatment plants.
- Conducted technical reviews of vendor offers for all electro-mechanical items within pumping stations and sewage treatment plants, ensuring all vendor queries were resolved and securing final client approval before procurement.

Responsibilities undertaken

Sep 2023 to present – Mentor to the project associates for the live project funded by DST, Government of India and IC-Impacts, Canada titled, " A Data-driven Simulation-based Machine Learning Optimization SML-Opt Framework for Net Zero Energy Building Retrofits."

Oct 2023 - Website coordinator of CTaM department

Handling the official CTaM department website using tools like GitHub, Win SCP, Atom, which has 80+ students enrolled, Strategized Plan of Action every week to update the Official department Website, Working in a team of 3+ coordinators, ensuring smooth operation of the updated website for entire CTaM department.

Jun 2018 – Estimation lead of Process/Mechanical team for various STP tenders/proposals

Sep 2016 – Design lead of Mechanical, Utility and in-plant piping team for various operating STP projects

Mar 2012 - Organized a workshop on Ornithopter Design and Fabrication at Kurukshetra (national level technical symposium organized by College of Engineering Guindy under the patronage of UNESCO).

- First of its kind in South India
- Targeted at school students, the workshop witnessed a footfall of well over five hundred, over the course of two days.
- 13 Mar 2012 – Concepts pertaining to design basis and fabrication of Ornithopter clearly explained to the students by self. Technical Lecture sessions were succeeded/followed by interactive Q&A sessions.
- 14 Mar 2012 - Hands-on training for the participants in making Ornithopter on their own by providing the necessary kits and guidance. Around 50 coordinators and volunteers were deployed for the same.
- Stationery and gifts awarded to the students with outstanding Ornithopter design.

Co-Curricular Industrial Experience

Jul 2013 to December 2013 - Design and fabrication of reactor to extract oil from waste plastic carried out under Dr. D Mohanlal, Professor & Head R&AC, Department of Mechanical engineering, College of Engineering Guindy.

- Designed SS pipe (pressure vessel) referring ANSI B36.19 and ASME code as standards for pipe Schedule, design of end cap, flange selection based on the working pressure.
- Utilized PID controller with thyristor drive for temperature control monitored using thermocouple sensor.

Vocational Training

15 to 20 Feb 2016 - Supervisory Development Program organized by L&T for future executives and managers.

- Batch consisted of 23 members who were nominated from approximately 200 peers.
- Topics covered: functional (eg: Tendering and Contracts, Working Capital Management, Indirect Taxation, etc.) and behavioral (eg: Interpersonal Effectiveness, Negotiation Skills, etc.).

April to June 2015 - Online courses on “Business Communication” and “Business Etiquette” organized by L&T in collaboration with Anytime Learning (ATL) India.

4 to 25 Sep 2014 - Program on Engineering, Technology and Business operations organized at Centre for Technology and Engineering Applications, Mysore.

24 Jun to 6 Jul 2013 - Short – term certificate course in CFD (Computational Fluid dynamics) using ANSYS – FLUENT, organized by AU – FRG institute for CAD/CAM, Anna University Chennai

10 to 22 Jun 2013 - Short – term certificate course in ADAMS (Automated dynamic Analysis of Mechanical Systems), organized by AU – FRG institute for CAD/CAM, Anna university Chennai.

Co-Curricular/Sports- Awards and recognition

March 2023 – Team Championship – Intra department Chess tournament – IIT Bombay

Feb 2015 & Feb 2016 – Champion - Inter-IC Club Chess tournament (for 2 consecutive years) - represented Water and Effluent Treatment IC.

Apr 2013 – Finished fourth / Second Semi-finalists - National level Inter College Team Chess Championship held at SSN College, Chennai. – represented College in the capacity of Team Leader.

Professional Profiles

Website: https://omprakashrr.github.io/portfolio/

LinkedIn: https://www.linkedin.com/in/omprakashrethnnam

Google Scholar: https://scholar.google.com/citations?user=lfaK8TwAAAAJ&hl=en